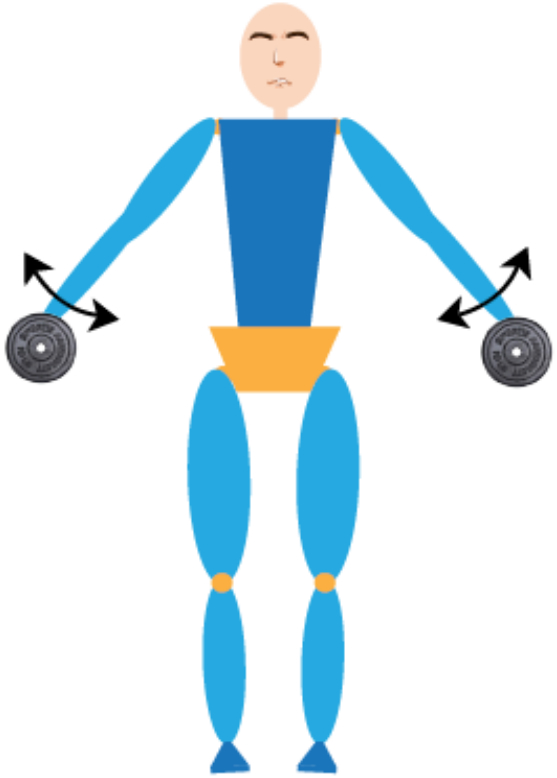
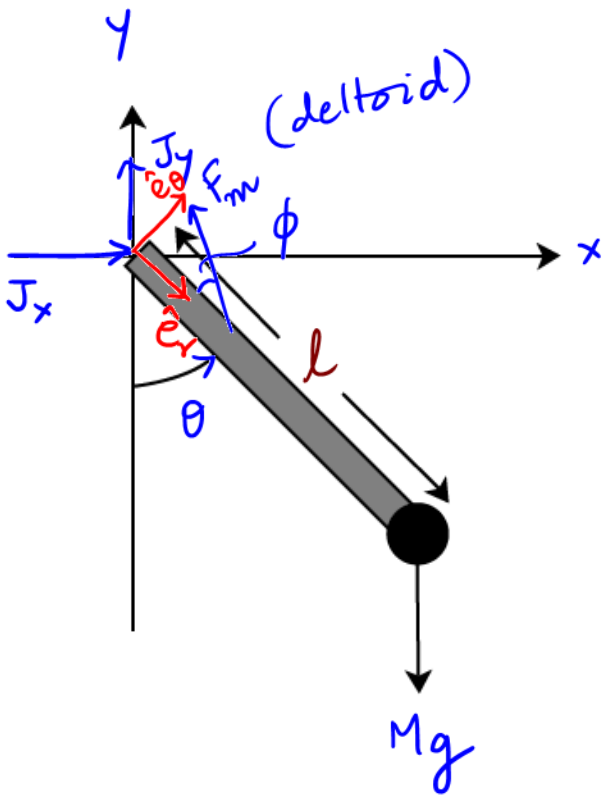
Angular motion

Arm is assumed to be weightless, and of length l

$M \rightarrow$ mass of dumbbell

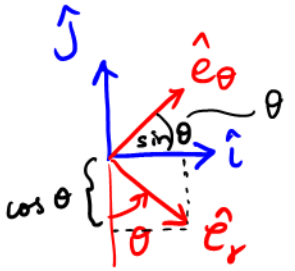




unit vector along

$\hat{e}_r$ – radial direction

$\hat{e}_\theta$ – tangential direction



$$\hat{e}_r = \sin\theta\,\hat{i} - \cos\theta\,\hat{j}$$

$$\hat{e}_\theta = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$$

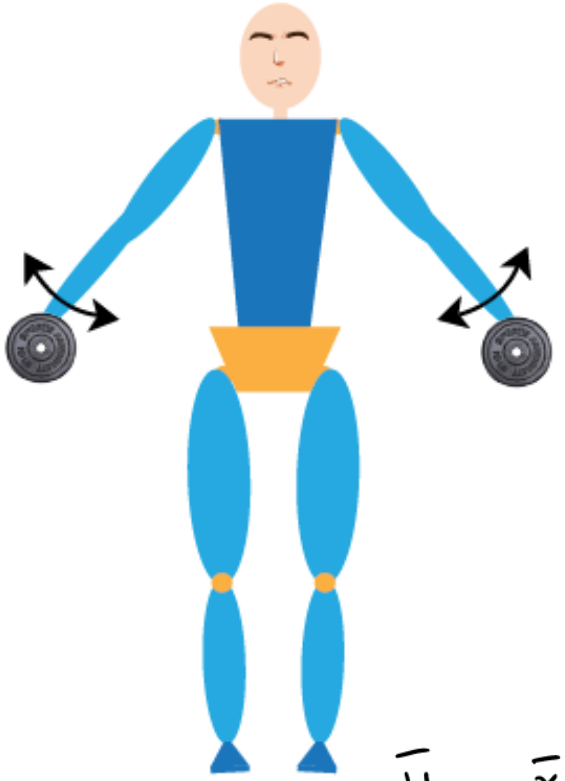
$$\frac{d\hat{e}_r}{dt} = \dot{\hat{e}}_r = \cos\theta\,\dot{\theta}\,\hat{i} + \sin\theta\,\dot{\theta}\,\hat{j} = \dot{\theta}\,\hat{e}_\theta$$

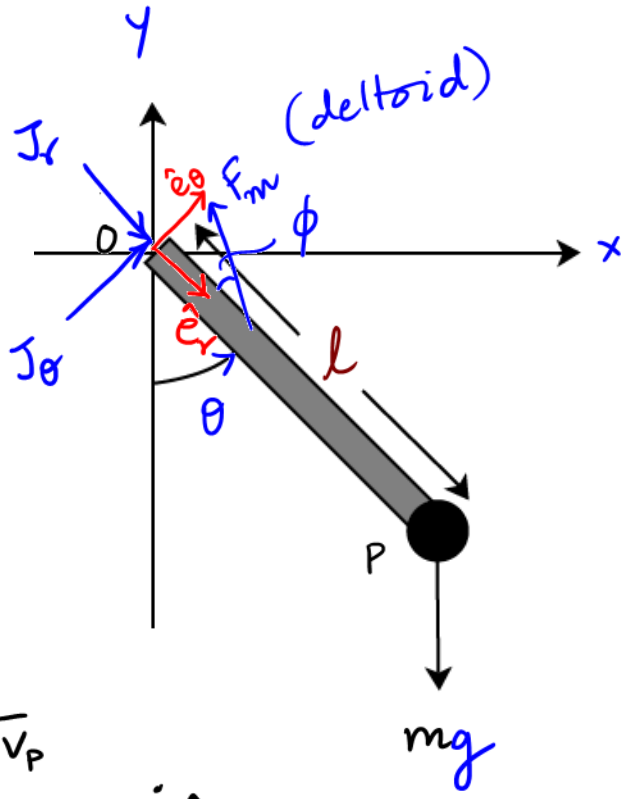
$$\frac{d\hat{e}_\theta}{dt} = \dot{\hat{e}}_\theta = -\sin\theta\,\dot{\theta}\,\hat{i} + \cos\theta\,\dot{\theta}\,\hat{j} = -\dot{\theta}\,\hat{e}_r$$

Angular motion

Arm is assumed to be weightless, and of length l

$m \rightarrow$ mass of dumbbell





$$\bar{r}_{P/O} = l\,\hat{e}_r$$

$$\bar{v}_P = \frac{d\bar{r}_{P/O}}{dt} = l\frac{d\hat{e}_r}{dt} = l\dot{\theta}\,\hat{e}_\theta$$

$$\bar{a}_P = \frac{d\bar{v}_P}{dt} = l\ddot{\theta}\,\hat{e}_\theta + l\dot{\theta}\,\dot{\hat{e}}_\theta$$

$$= l\ddot{\theta}\,\hat{e}_\theta - l\dot{\theta}^2\,\hat{e}_r$$

$\dot{\theta} \rightarrow \omega$ angular vel.

$\ddot{\theta} \rightarrow \alpha$ angular acceleration

$$\bar{H}_O = \bar{r}_P \times m\bar{v}_P$$

$$= l\,\hat{e}_r \times m\,l\dot{\theta}\,\hat{e}_\theta$$

$$= ml^2\dot{\theta}\,\hat{k}$$

$$\Sigma F_r = m a_r$$

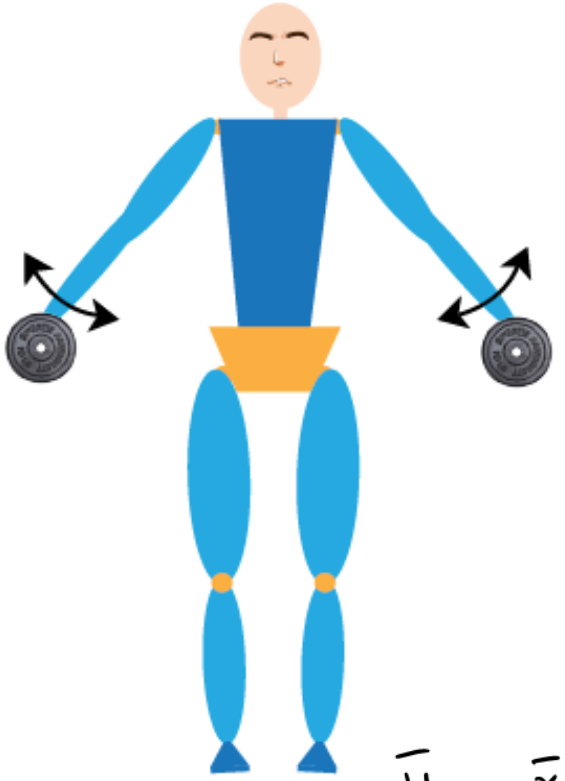
$$\Sigma F_\theta = m a_\theta$$

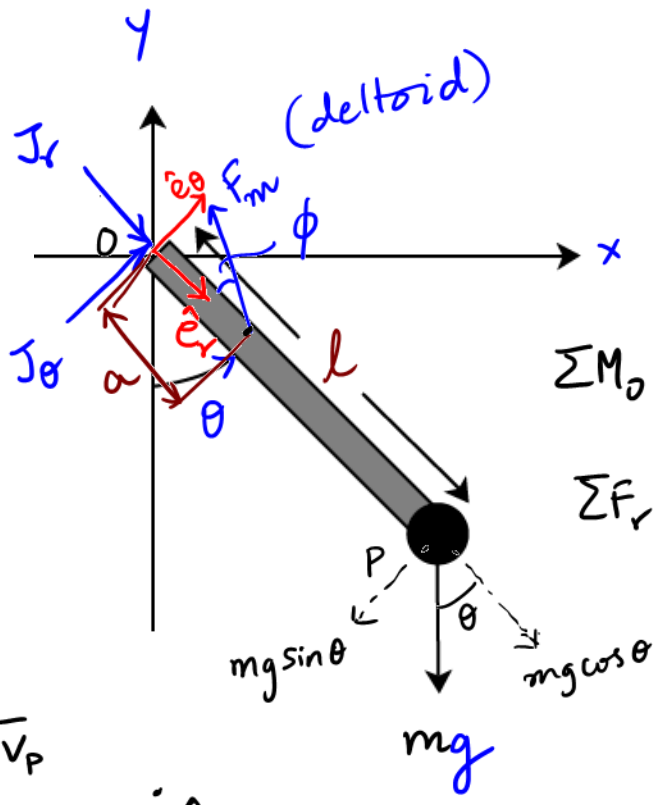
$$\Sigma M_O = \frac{d\bar{H}_O}{dt}$$

Angular motion

Arm is assumed to be weightless, and of length ℓ

$m \rightarrow$ mass of dumbbell





$$\frac{d\bar{H}_o}{dt} = \underbrace{m\ell^2}_{I_o}\underbrace{\ddot{\theta}}_{\alpha}\hat{k}$$

Knowing $\theta = \theta(t)$, I can use these 3 eqns to solve for J_r, J_θ, F_m

$$\Sigma M_o: \quad F_m \sin\phi \cdot a - mg\sin\theta \cdot \ell = m\ell^2\ddot{\theta}$$

$$\Sigma F_r: \quad J_r + mg\cos\theta - F_m\cos\phi = -m\ell\dot{\theta}^2$$

$$\Sigma F_\theta: \quad J_\theta - mg\sin\theta + F_m\sin\phi = m\ell\ddot{\theta}$$

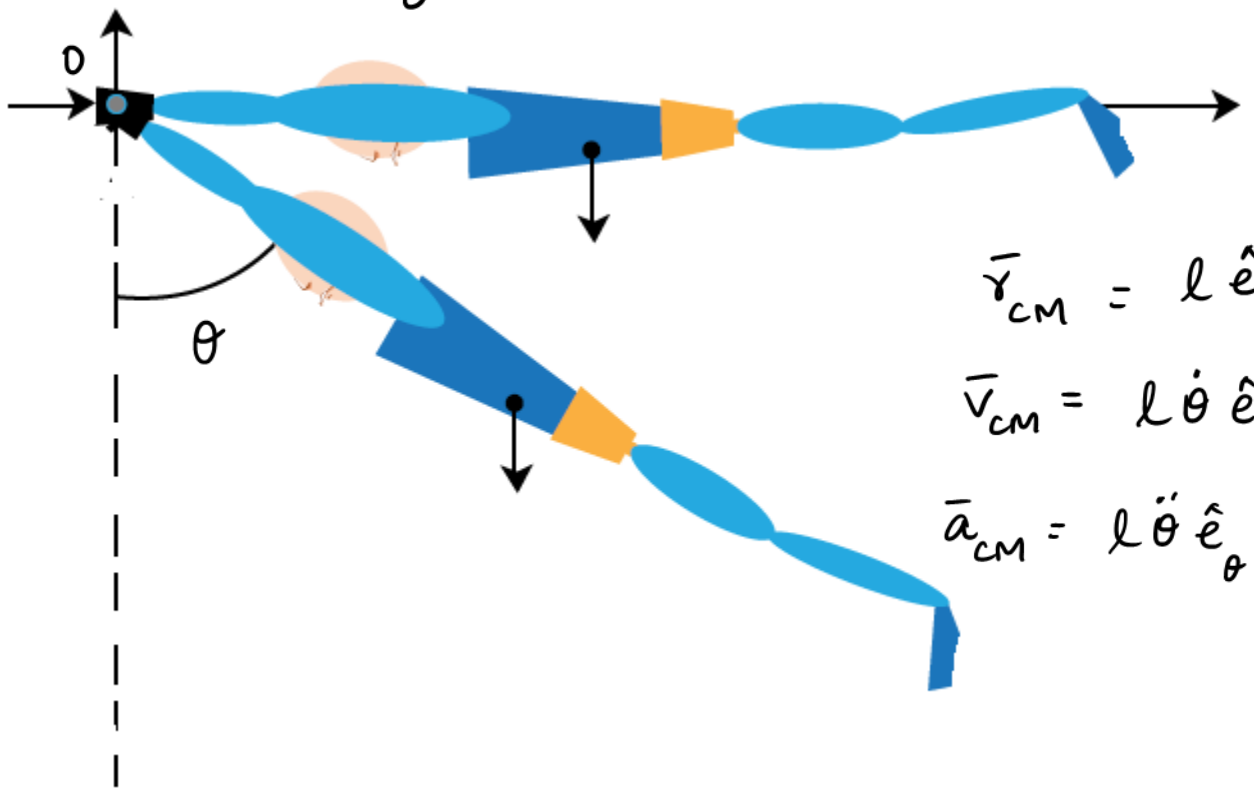
$$\bar{H}_o = \bar{r}_p \times m\bar{v}_p$$

$$= \ell\hat{e}_r \times m\ell\dot{\theta}\hat{e}_\theta$$

$$= m\ell^2\dot{\theta}\hat{k}$$

Angular motion – distributed mass

Represent gymnast by rod of uniformly distributed mass m, length 2ℓ



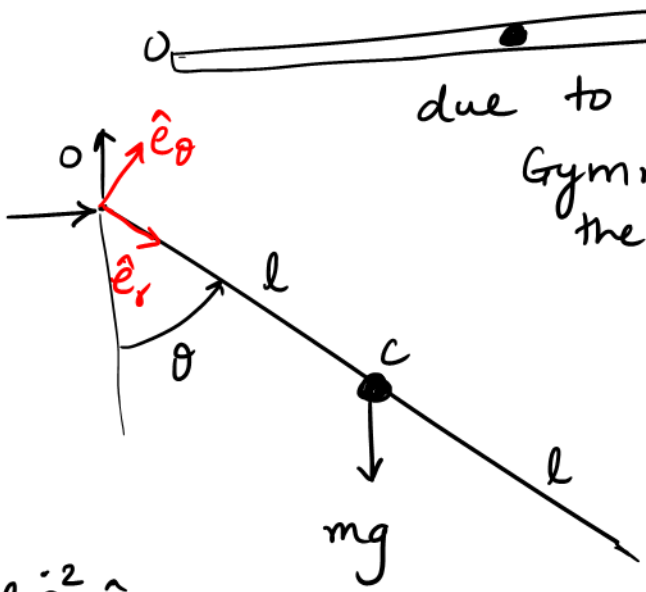
$$\bar{r}_{CM} = \ell \hat{e}_r$$

$$\bar{v}_{CM} = \ell \dot{\theta} \hat{e}_\theta$$

$$\bar{a}_{CM} = \ell \ddot{\theta} \hat{e}_\theta - \ell \dot{\theta}^2 \hat{e}_r$$

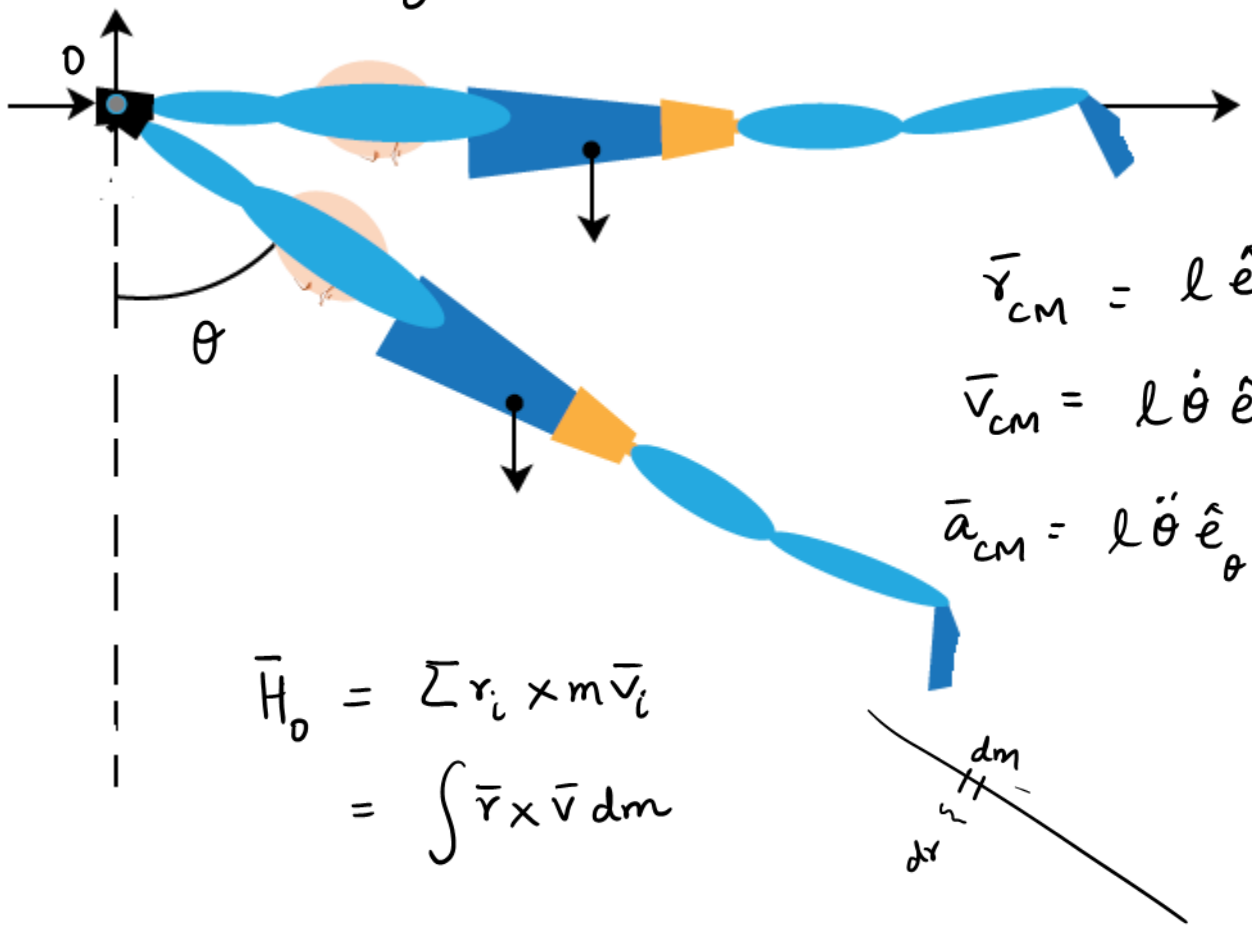
Wrist acts as a pin joint as the gymnast rotates around the bar due to gravity.

Gymnast let go from the hor. position



Angular motion – distributed mass

Represent gymnast by rod of uniformly distributed mass m, length 2ℓ



$$\bar{r}_{CM} = \ell \hat{e}_r$$

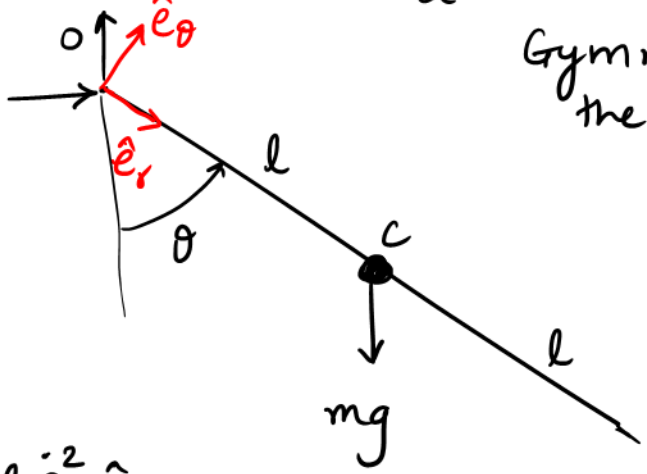
$$\bar{v}_{CM} = \ell \dot{\theta} \hat{e}_\theta$$

$$\bar{a}_{CM} = \ell \ddot{\theta} \hat{e}_\theta - \ell \dot{\theta}^2 \hat{e}_r$$

$$\bar{H}_O = \sum r_i \times m \bar{v}_i = \int \bar{r} \times \bar{v}\, dm$$

Wrist acts as a pin joint as the gymnast rotates around the bar due to gravity. Gymnast let go from the hor. position



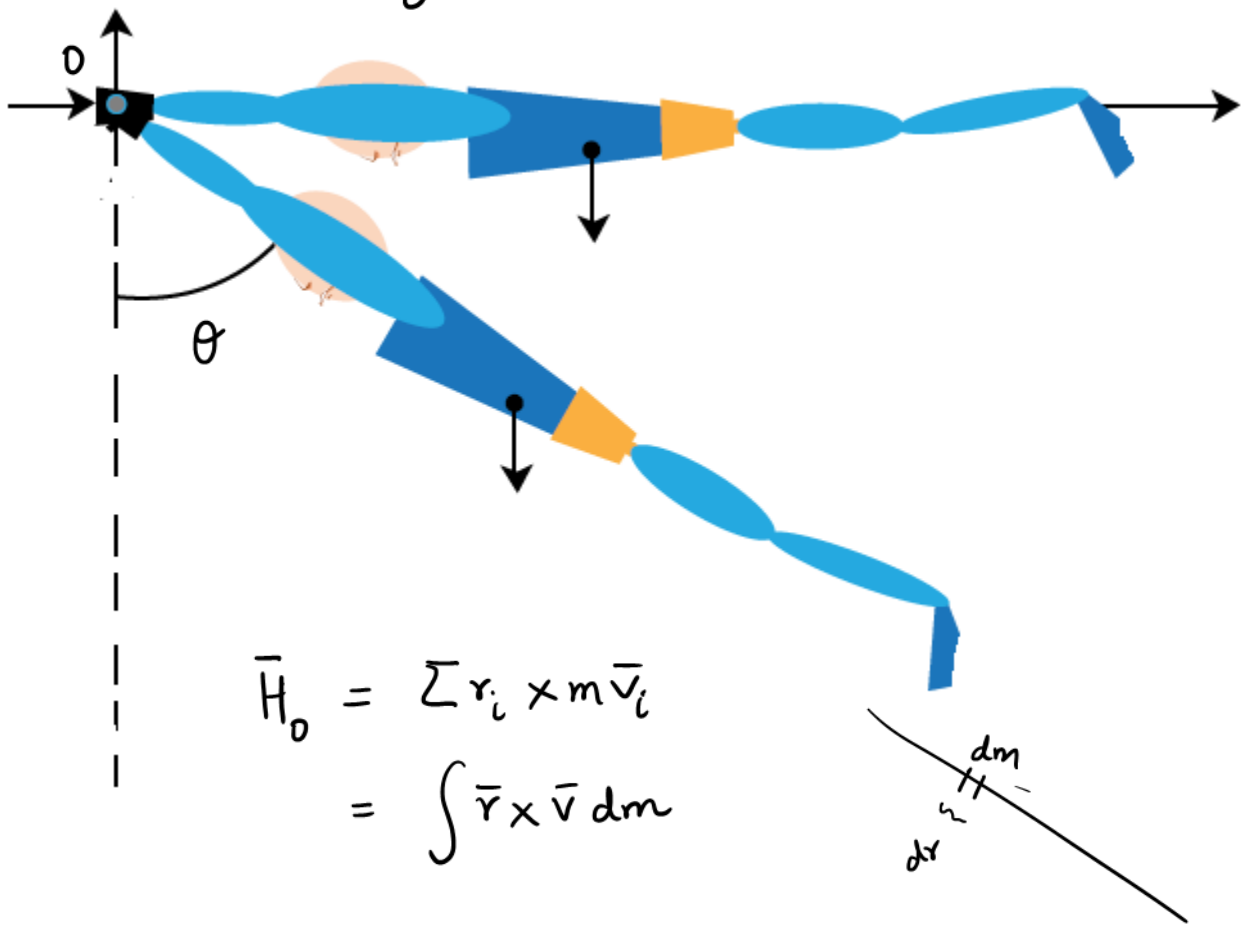


Rod of length 2ℓ has mass m

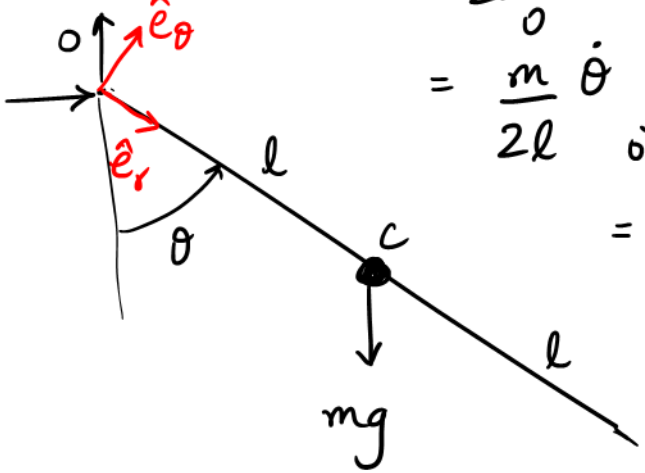
$$dm = \frac{m}{2\ell} dr$$

Angular motion – distributed mass

Represent gymnast by rod of uniformly distributed mass m, length 2ℓ



$$\bar{H}_O = \sum \bar{r}_i \times m\bar{v}_i = \int \bar{r} \times \bar{v}\, dm$$



$$\bar{H}_O = \int_0^{2\ell} r\hat{e}_r \times r\dot{\theta}\hat{e}_\theta \frac{m}{2\ell} dr$$

$$H_O = \frac{m}{2\ell}\int_0^{2\ell} r^2\dot{\theta}\, dr$$

$$= \frac{m}{2\ell}\dot{\theta}\int_0^{2\ell} r^2 dr$$

$$= \frac{m}{2\ell}\dot{\theta}\,\frac{r^3}{3}\Big|_0^{2\ell}$$

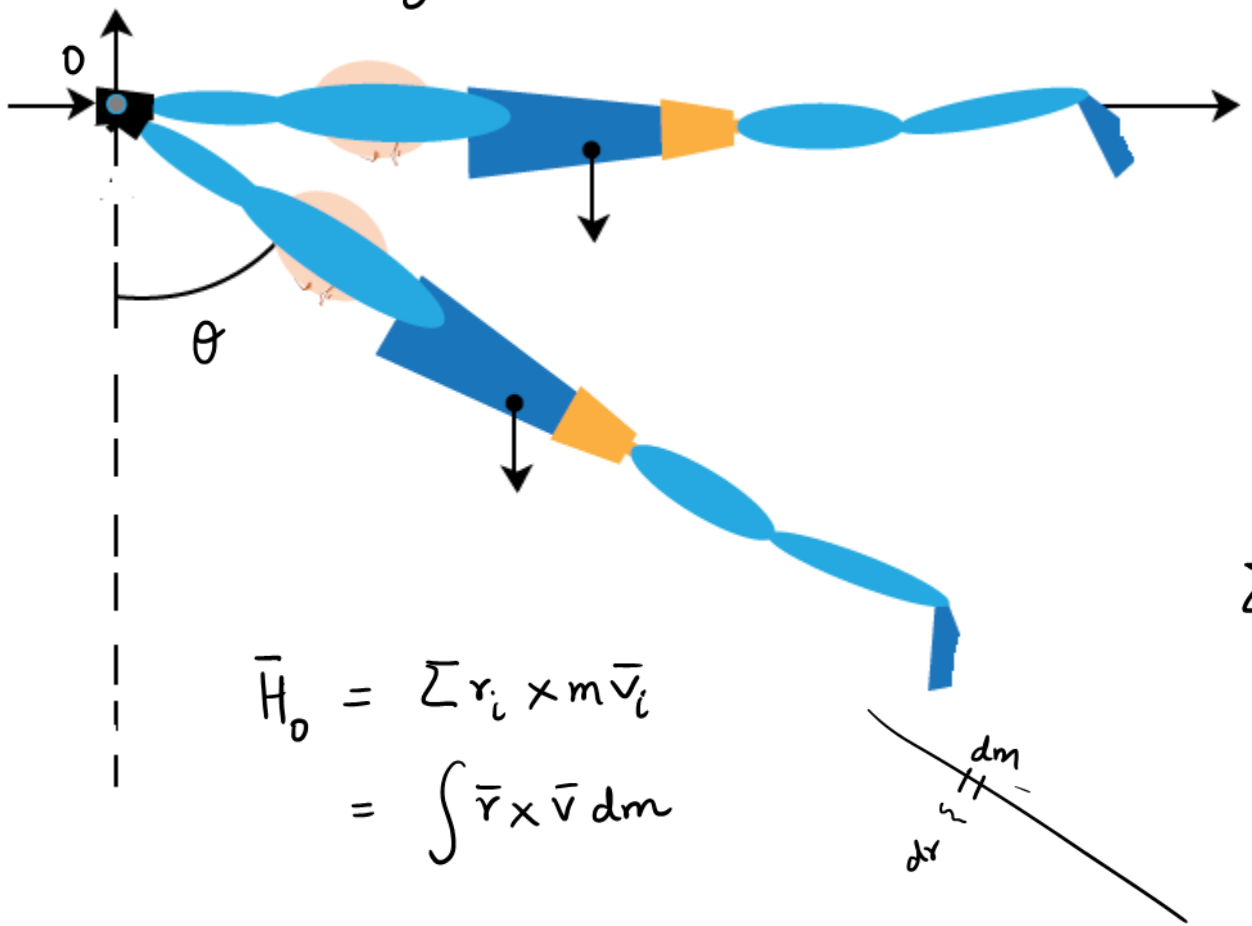
$$= \frac{4m\ell^2\dot{\theta}}{3}$$

Rod of length 2ℓ has mass m

$$dm = \frac{m}{2\ell} dr$$

Angular motion – distributed mass

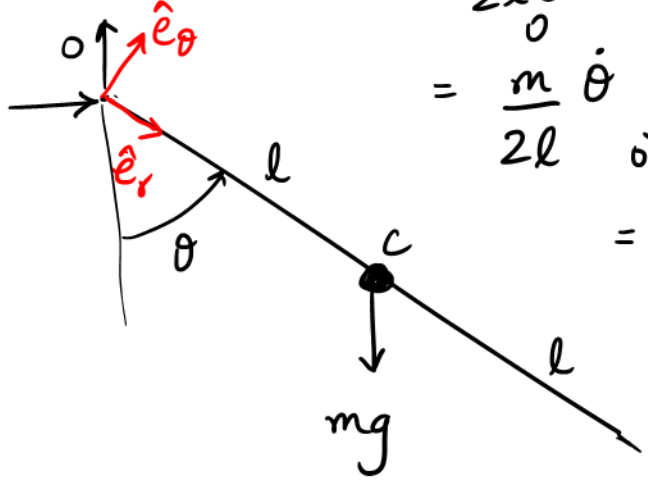
Represent gymnast by rod of uniformly distributed mass m, length 2ℓ



$$\bar{H}_O = \sum \bar{r}_i \times m\bar{v}_i = \int \bar{r} \times \bar{v}\, dm$$

$$\bar{H}_O = \int_0^{2\ell} r\hat{e}_r \times r\dot{\theta}\hat{e}_\theta \frac{m}{2\ell} dr$$

$$H_O = \frac{m}{2\ell}\int_0^{2\ell} r^2\dot{\theta}\, dr = \frac{m}{2\ell}\dot{\theta}\int_0^{2\ell} r^2 dr = \frac{m}{2\ell}\dot{\theta}\,\frac{r^3}{3}\Big|_0^{2\ell} = \frac{4}{3}m\ell^2\dot{\theta}$$

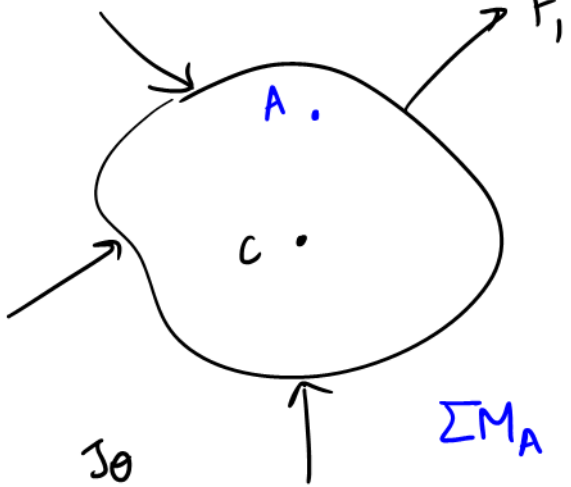


$$\Sigma M_O = \frac{dH_O}{dt} = \frac{4}{3}m\ell^2\ddot{\theta} = I_O\alpha$$

$$I_C = \frac{1}{12}m(2\ell)^2 = \frac{m\ell^2}{3}$$

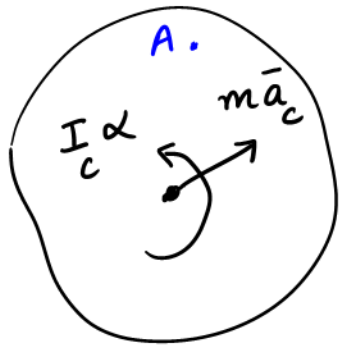
$$I_O = I_C + m\ell^2 = \frac{m\ell^2}{3} + m\ell^2 = \frac{4}{3}m\ell^2$$

Parallel axis theorem

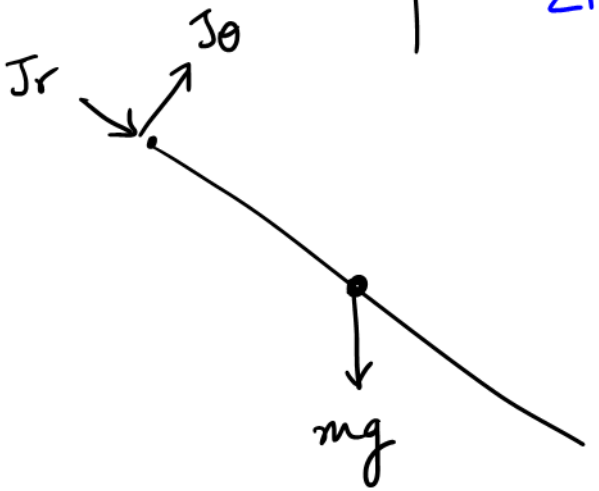


=

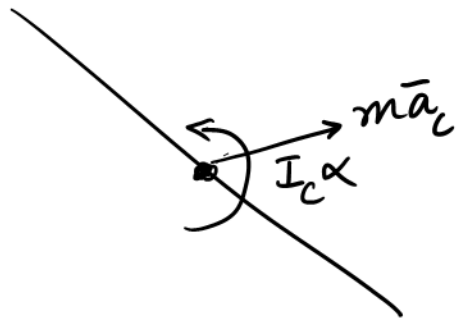
Kinetic diagram



$= (\Sigma M_A)_K$



=



$$\Sigma \bar{F} = m\bar{a}_C$$

$$\Sigma M_C = I_C\alpha$$

$$\bar{a}_C = l\ddot{\theta}\,\hat{e}_\theta - l\dot{\theta}^2\hat{e}_r$$

$$I_C = \frac{m}{12}(2l)^2 = \frac{ml^2}{3}$$